# Alexis J. Diamond, Ph.D.


CONTACT

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PROFILE

Quantitative methodologist and hands-on builder. I trained under Donald Rubin and, with Alberto Abadie and Jens Hainmueller, co-authored the foundational papers on the **synthetic control method** (*JASA* 2010; *AJPS* 2015) and its ***Synth*** software (**200,000+ cumulative CRAN downloads**). The method is now a standard tool for estimating causal effects where experiments are impossible. My first research work was on **catastrophic risk**: at RAND I built statistical early-warning systems for bioterror detection and ran structured adversarial tests of whether public-health agencies could actually respond in time. Over the past two years, as Minerva's first **AI Fellow**, I have built AI-enabled tools for professors and students at my university. I am now applying this background to **AI safety**, where many of the hardest open problems are measurement problems: whether an evaluation measures what it claims to, and whether a model's stated reasoning explains its behavior. My current paper, *Who Took the Hint?* (Agent Evaluation Science symposium, New York, November 2026), applies causal inference to **chain-of-thought monitorability** evaluations: it shows they have the structure of an encouragement design, and that the standard monitorability score grades monitors against a different quantity from the one they are meant to detect, so a correctly computed score can support the wrong conclusion. I also serve as secondary advisor to nine student **AI safety capstone** projects.

SOFTWARE & SYSTEMS BUILT

***Synth*** **— R package for synthetic control methods** *2011–present*
*Co-author* (with Abadie & Hainmueller) — **200,000+ cumulative CRAN downloads**; the reference implementation of the synthetic control estimator, used across economics, political science, epidemiology, and public health. Published in the *Journal of Statistical Software* (2011).

**rctvsobs.org — causal-inference benchmark portal** *2026*
*Co-founder & co-builder* — open portal curating **paired RCT/observational benchmarks** (LaLonde and others), so that estimators can be scored against known experimental ground truth. The first online resource of its kind.

**LLM systems designed and shipped solo** *(LLM-assisted development)* *2024–present*

- **SupaNerd** — adaptive assessment platform: generates items from arbitrary source material, schedules review (FSRS), interleaves practice across 10+ item types, and tracks per-student mastery. *TypeScript/Next.js, Anthropic Claude API, PostgreSQL.* In production use.
- **Coursebuilder Agent** — *human-in-the-loop* LLM agent that revises lesson plans grounded in real class transcripts (analyze → diagnose → draft → apply). The agent proposes changes for a human to approve rather than applying them itself, and shows every change as a reviewable diff. *Chrome extension (Manifest V3)*; in private faculty pilot.
- **GWiz** — rubric-aligned LLM grading assistant: structured evaluation and feedback across assignments and submissions, built to make model judgments auditable against an explicit rubric. *TypeScript/Next.js.*

**Class Transcriber** *2025*
*Team project* — scoped the problem and directed and supervised a student development team, who built a Whisper-based pipeline that turns class recordings into structured transcripts (PDF/CSV/JSON). *Python, local and privacy-preserving.*

**OpenAI** *July–September 2026*
*Alpha Tester* (volunteer) — tested and gave feedback on AI tools for educators (under NDA)

## Catastrophic Risk & Early Warning

**Minerva University — AI Safety Capstone** *2026–2027*
*Secondary Advisor* — nine student projects on reward hacking, evaluation integrity, deception detection, monitoring and unlearning

**AIxBio Africa** *2026*
*Facilitator* — Introduction to AI & Biosecurity: An African Perspective (Cohort 1); leading six live discussion sessions for a participant group

**RAND Corporation**, Arlington, Virginia *2004–2005*
*Summer Associate, then Consultant* — work for the U.S. Department of Health and Human Services on detection of and response to low-probability, high-consequence events

- Developed a statistical **early-warning system for bioterror response** in the Washington, D.C. area — detecting a rare adversarial event against a noisy baseline, where both false negatives and false alarms are costly
- **Served as the covert caller and data collector** on a structured test of emergency response: placed simulated urgent case-report calls to health departments in nineteen metropolitan areas, sustaining a credible clinician persona with expert subjects, and collected the response-time data against the CDC 24/7 standard (*Health Affairs*)
- Evaluated competing **statistical methods for syndromic surveillance** — i.e. which detection methods actually work on real data, and under what assumptions they fail
- Built preparedness training and assessment exercises adopted by local public health agencies

## Quantitative Research

**Minerva University**, San Francisco *2015–present*
*Professor, Computational Sciences; AI Fellow (2025–2026, first to hold the role)*

- Supervised **60+ student theses** over 10 years in data science, causal inference, machine learning, and impact evaluation — scoping tractable empirical questions and driving them to a written result on a fixed timeline
- Designed and taught undergraduate and graduate courses in statistics, computer science, logic, and microeconomics
- Wrote and shepherded Minerva's institution-wide **AI Use Policy**, adopted across all faculty

**German Federal Employment Agency / Datametrics** *2004–2006*
*Evaluation Consultant* — built statistical software (*TrEffer*) with Donald Rubin's team to evaluate Germany's multibillion-euro job-training programs: matching estimators at national scale, in production

**Audible.com** (Amazon subsidiary) *2017–2018*
*Impact Evaluation Design Consultant* — designed a randomized controlled trial to estimate the effect of audiobooks on children's literacy

*Recognition:* Two-time **Gosnell Prize** winner (APSA, 2005 & 2007); World Bank "Knowbel" Prize (2013 & 2023); Fulbright Fellow; Phi Beta Kappa.

## Technical Skills

- *Languages I write:* **R** (advanced — two decades; author of a published, widely used package; read, debug, and review others' code fluently); **Python** (intermediate)
- *LLM-assisted development:* directed coding models to build and ship **three applications** (TypeScript/Next.js, PostgreSQL, Chrome extensions) — scoping requirements, testing behavior against them, and iterating to working software in stacks I do not write by hand
- *LLM engineering:* designed and operate production systems on the Anthropic Claude and Google APIs; agentic and human-in-the-loop architectures; prompt engineering; systematic evaluation of my own shipped tools against rubrics and real classroom use
- *Measurement & inference:* experimental design (RCTs), psychometrics and construct validity, causal inference under confounding (synthetic control, genetic matching, propensity methods), meta-analysis, benchmark construction and validation

## Publications

*16,000+ citations (Google Scholar).*

*Peer-reviewed papers & book chapters*

- Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program (with Abadie & Hainmueller), *Journal of the American Statistical Association* (2010). *Gosnell Prize; among the journal's ten most-cited articles, 2009–2011.*
- Genetic Matching for Estimating Causal Effects: A General Multivariate Matching Method for Achieving Balance in Observational Studies (with Sekhon), *Review of Economics and Statistics* (2013). *Gosnell Prize.*
- Comparative Politics and the Synthetic Control Method (with Abadie & Hainmueller), *American Journal of Political Science* (2015).
- *Synth*: An R Package for Synthetic Control Methods in Comparative Case Studies (with Abadie & Hainmueller), *Journal of Statistical Software* (2011).
- Estimating Poverty Rates in Subnational Populations of Interest: An Assessment of the Simple Poverty Scorecard (with Skoufias et al.), *World Development* (2020).
- Public Health Response to Urgent Case Reports: A Structured Test Reveals Gaps in Response Times in Nineteen Communities Across the Country (with Dausey & Lurie), *Health Affairs* (2005).
- Evaluating the Community DentCare Network in Northern Manhattan (with R. Diamond, Litwak & Marshall), *Journal of Public Health Dentistry* (2003).
- Evaluating Statistical Methods for Syndromic Surveillance (with Stoto et al.), in Wilson, Wilson & Olwell (eds.), *Statistical Methods in Counterterrorism*, Springer (2006).

*Institutional technical reports*

- Bioterrorism Preparedness Training and Assessment Exercises for Local Public Health Agencies, RAND Corporation Technical Report (2005).
- Tests to Evaluate Public Health Disease Reporting Systems in Local Public Health Agencies, RAND Corporation (2005).
- Women's Economic Activity and Engagement in the Informal Sector in Georgia, UN Women and the Swiss Agency for Development and Cooperation (2018).
- Women's Economic Activity and Engagement in the Informal Sector in Armenia, UN Women and the Swiss Agency for Development and Cooperation (2018).

*Working & discussion papers*

- Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program, NBER Working Paper / Harvard University (2007, 2009).
- Genetic Matching for Estimating Causal Effects, Harvard Kennedy School Faculty Research Working Paper / UC Berkeley (2005–2011).
- Estimating Poverty Rates in Target Populations: An Assessment of the Simple Poverty Scorecard and Alternative Approaches, World Bank Policy Research Working Paper No. 7709 (2016).
- Poverty Literature Review: Microfinance and Poverty Reduction, World Bank Group Working Paper (2012).

## Education

**Harvard University** — Ph.D., Political Economy & Government, 2008
*Supervisor:* Donald B. Rubin (John L. Loeb Professor of Statistics). *Dissertation:* Reliable Causal Inference in Non-experimental Settings

**Carleton University** — M.A., International Affairs, 1999 (Canada–US Fulbright Fellow)
**Alfred University** — B.A., Economics (Honors) & Mathematics, 1997 (Phi Beta Kappa)

*Additional training:* **BlueDot Impact** — Technical AI Safety (completed Sept. 2026); Technical AI Safety Project Sprint (in progress). **GEAR** — Gradual Empowerment Alignment Research, ARENA-based course (in progress, Oct.–Dec. 2026). **Lens Academy** — Compute Verification 1: technical and governance challenges of verifying international AI agreements (in progress).

## Evaluation Experience

*Two decades measuring impacts at scale for international institutions.*

**International Finance Corporation (IFC), World Bank Group** *2006–2024*
*Impact & Performance Evaluation Global Lead, Washington D.C. (2011–2015); Regional M&E Officer, Hanoi (2009–2011); M&E Officer (2006–2009); Consultant (2017–2024)*

- Led IFC's global impact- and performance-evaluation strategy and served as technical lead on all evaluations and research studies, setting the methodological standard across the institution
- Managed evaluation officers across six country offices; trained hundreds of staff worldwide on measurement and evaluation
- As consultant, quality-assured and led impact evaluations across 30+ countries — Bangladesh economic zones, a sub-Saharan Africa agriculture index-insurance study (technical lead), Indonesia MSME finance, Eswatini trade facilitation — and advised on applications of the synthetic control method

**International Fund for Agricultural Development (IFAD)** *2023–2025*
*Senior Evaluation Consultant, Independent Office of Evaluation* — methods expert on a corporate-level evaluation of IFAD's Impact Assessment System, the evidence base for its results reporting to donors

- Tested the project sampling frame statistically for representativeness across three replenishment cycles
- Systematically reviewed a 25% sample of impact assessments, with **full replication of code and data** in eight deep-dive cases
- Assessed the aggregation of results for corporate reporting against meta-analysis principles; synthesized findings into a rated evaluation report with recommendations

**Caucasus Research Resource Center (CRRC)**, Tbilisi, Georgia *2016–2018*
*Impact Evaluation Consultant & Principal Investigator* — co-led a quasi-experimental evaluation of a major IFAD-funded agriculture, infrastructure, and finance program; senior economist for UN Women–sponsored studies of female economic activity in Georgia and Armenia

**Asian Development Bank**, Manila *2018–2019*
*Evaluation Consultant* — developed and piloted an ex-post assessment protocol with the Independent Evaluation Group; assessed project completion reports against a complex rubric; designed M&E training for advisory staff

**Dalberg Global Development Advisors** *2017*
*Impact Evaluation Design Consultant* — advised on quasi-experimental design for a Google/USAID broadband program; delivered statistical-methods training to staff

## Visiting Appointments

**Central European University**, School of Public Policy — Visiting Chair, Vienna (2025); Visiting Professor, Budapest (2015–2017)
*Taught machine learning, results measurement, and program evaluation to MA and PhD students; advised on quantitative methods.*

**Harvard University**, John F. Kennedy School of Government — Visiting Professor (2013)
*Taught Impact Evaluation (ECON128), satisfying the PhD econometrics requirement.*

## Selected Talks

- *Presenter*, "Augmentation, Not Automation: AI-Enabled Rigor and Access Across Disciplines," WSCUC ARC2026, San Francisco (2026)
- *Featured Speaker*, "AI, Jobs, and Education," Root Impact, Seoul (2025)
- *Public Talk*, "Age of AI," National Taiwan University Design & Innovation School (2025)
- *Featured Speaker*, "Lean Impact Evaluation," Mingyi Foundation, Taiwan (2025)
- *Featured Speaker*, Amazon.com Friday Economics Seminar, Seattle (2019)
- *Presenter*, World Bank Results Forum, Washington D.C. (2019)
- *Keynote*, Methodological Conference on Research for Development, Tbilisi (2017)
- *Presenter*, American Evaluation Association Annual Conference, Denver (2014)
- *Invited Talk*, Department for International Development (DFID), London (2012)
- *Invited Talk*, Advanced Statistics Workshop, Harvard University (2004, 2006)
- *Invited Talk*, Econometrics Workshop, MIT (2005)
- *Presenter*, Political Methodology Summer Conference, Florida State University (2005)

*Languages:* English (native); Spanish (conversational; translator, 1999 Pan American Games)

*International:* Professional work in 30+ countries across Africa, Asia, Europe, and Latin America; lived in Hanoi (2009–2011) and Budapest (2015–2017)

*Citizenship:* Dual citizen — Canada and the United States.